SoC Occupancy Aging and Energy–Reserve Bidding Under Price Stress: A Controlled SMARD-Based Diagnostic

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***Abstract*—Battery owners who sell energy and reserve simultaneously must park a portion of their stored energy to cover upward reserve commitments. Throughput and depth-of-discharge cost models do not charge for this parked energy, even though calendar aging depends strongly on the state of charge. This paper asks when charging for it actually changes a bid. A virtual power plant with a 2.5 MW/5 MWh battery is modeled in one daily mean–CVaR linear program, and three aging models are swapped inside it: a throughput penalty, a piecewise-linear daily depth model, and a reserve-aware layered calendar–cycle model (ReLCCA). We use public SMARD data for Germany/Luxembourg (262 complete days, December 2025–August 2026); reserve and imbalance prices are documented proxies. On ordinary days the three models agree. On 1 May 2026, when the day-ahead price fell to -499 EUR/MWh at midday, the throughput model's energy-only bid went long while ReLCCA's stayed short. The two energy-only books differ by 1,312 EUR; 1,246 EUR of that is the day-ahead position and only 4 EUR is the direct aging cost. Putting the same price trough on another summer day did not reproduce the reversal, and a scan of 20 negative-price days found no second case. We read this as a mechanism that needs both price stress and the right portfolio state, not as a ranking of aging models. Because reserve and imbalance prices are proxies, all euro values are simulated, not realized, settlements.**

***Index Terms*—Battery energy storage, battery degradation, calendar aging, CVaR, energy–reserve bidding, linear programming, SMARD, virtual power plant.**

I. Introduction

VIRTUAL power plants (VPPs) aggregate distributed energy resources and expose a heterogeneous portfolio to electricity markets through coordinated scheduling [1]–[4]. Their short-term value depends on the interaction between uncertain renewable production, load, market prices, storage flexibility, and ancillary-service commitments. This has motivated stochastic, robust, and risk-aware bidding models for VPPs and storage resources [5]–[8].

For battery energy storage systems (BESSs), however, the economic value of flexibility is inseparable from degradation. Battery aging is commonly separated into cycle aging and calendar aging. Experimental and semi-empirical studies have shown that degradation depends on operating depth, state of charge (SoC), temperature, time, and electrochemical conditions [9]–[14]. Market-oriented optimization has consequently adopted throughput costs, cycle-depth functions, and more detailed aging representations to maintain computational tractability [15]–[20].

A specific modeling issue arises when the same BESS is used for energy arbitrage and reserve capacity. An upward reserve commitment requires sufficient energy to remain available for deployment. Thus, some stored energy is deliberately parked rather than cycled. If calendar aging depends on SoC, the economic exposure of this parked energy is not represented by a pure throughput penalty. The resulting effect is potentially indirect: a small change in the degradation term can alter the marginal value of stored energy and, through the optimization constraints, change the preferred market position.

The objective of this study is not to claim that calendar-aware degradation is universally superior to cycle-centric formulations. Instead, the paper asks a narrower question: under what market and residual-portfolio conditions does explicit SoC-occupancy pricing change the optimal mean–CVaR energy–reserve position? The question is examined through a paired design in which the plant, scenario inputs, market assumptions, risk parameters, and solver remain fixed while the aging representation

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The contributions are fourfold. First, a common linear market formulation allows three aging representations to be exchanged without changing the surrounding optimization model. Second, ReLCCA combines SoC-occupancy and band-flow costs with an explicit upward-reserve parking constraint. Third, a profit-decomposition procedure separates direct aging expenditure from the downstream change in the optimized day-ahead energy position. Fourth, a controlled negative-price intervention and a scan of qualifying days distinguish a price-only explanation from a conditional portfolio-state mechanism. The remainder of the paper is organized as follows. Section II reviews related work and positions the contribution. Section III describes the VPP, data, information structure, and market proxies. Section IV gives the linear formulation. Section V defines the paired experimental protocol. Section VI presents the numerical results. Section VII discusses the mechanism and its interpretation. Section VIII states limitations and reproducibility requirements. Section IX concludes the paper.

II. Related Work and Research Gap

The literature relevant to this study can be organized into four strands: VPP market participation, uncertainty and risk, battery degradation, and energy–reserve co-optimization. Early VPP work established aggregation and coordinated market participation as an operational paradigm [1], [3], [4]. Stochastic and robust formulations subsequently addressed uncertainty in renewable output and prices, including price-taker and adaptive-robust offering models [5]–[7], [21]. Risk-averse energy and reserve scheduling has also been studied explicitly for VPPs [8].

The second strand concerns battery aging. Foundational reviews and experiments distinguish calendar and cycle mechanisms and document the strong dependence of calendar aging on SoC [9]–[14]. Optimization studies have translated these relationships into computational models. Hesse *et al.* use mixed-integer linear optimization for aging- and efficiency-aware arbitrage [15]. Xu *et al.* develop market-oriented cycle-aging cost formulations [16], [17], while Shi *et al.* analyze cycle-aging mechanisms in pay-for-performance settings [18]. More recent work considers convex-hull and electro-thermal representations to balance aging fidelity and optimization complexity [19], [20].

More recent aging-aware operation studies reinforce the need to distinguish a tractable dispatch penalty from a validated electrochemical lifetime model. A review by Collath *et al.* organizes the principal aging stressors and the ways degradation is embedded in battery operation [22]. Calendar-aging relationships are chemistry-dependent and need not be monotonic across the full SoC range; time dependence is also often represented by a sublinear law, whereas optimization models may adopt simpler linear approximations for tractability [12], [14], [22]. In an intraday-arbitrage study, Collath *et al.* show that linearized calendar and cycle-aging representations can materially change lifetime-oriented scheduling and profitability relative to simpler throughput-based treatment [23]. The implication for the present work is methodological rather than quantitative: SoC residence can affect a market decision even when the aging law is simplified, but the coefficients and resulting economic effects remain conditional on the chemistry, calibration data, and operating assumptions. ReLCCA is therefore treated here as an optimization-oriented occupancy-cost representation, not as a cell-level capacity-fade predictor.

The third strand concerns reserve participation. Storage has been co-optimized across energy and reserve products, including formulations with probabilistic delivery guarantees and German secondary-control-reserve applications [24]–[27]. These studies establish that reserve commitments impose operational and opportunity-cost constraints. Related wind–battery VPP models incorporate degradation and reserve scheduling jointly [27], [28]. Studies of frequency-reserve operation also show why reserve capacity and reserve activation should not be conflated. Aaltonen *et al.* evaluate market bidding with a physics-based aging model in a reinforcement-learning framework for frequency-containment reserve [29], while Astero and Evens examine battery operation under FCR technical requirements [30]. These works complement energy–reserve co-optimization studies, but they do not remove the need to represent activation energy, signal mileage, and SoC recovery when those quantities are available. In the present study, reserve is represented through capacity revenue and a fixed energy-parking constraint; activation trajectories are not simulated.

TABLE I
COMPARISON OF REPRESENTATIVE STUDIES AND POSITIONING OF THE PRESENT WORK

Study	Aging	Risk / uncertainty	Energy–reserve	Relevance
Hesse *et al.* [15]	Aging + efficiency	Deterministic	Energy arbitrage	Optimization-aware degradation
Xu *et al.* [16], [17]	Cycle depth	Market uncertainty	Energy / reserve	Market-oriented cycle cost
Shi *et al.* [18]	Cycle aging	Operational uncertainty	Regulation	Cycle-aging control
Toubeau *et al.* [24]	—	Probabilistic	Energy / reserve	Reserve delivery
Merten *et al.* [25], [26]	Battery-aware	Market forecasting	aFRR	German reserve participation
Fernández-Muñoz and Pérez-Díaz [27]	DoD / rate	Stochastic	DA + secondary reserve	Wind–battery VPP
Ghanaee *et al.* [28]	Calendar + cycle	Deterministic	DA + reserve	Recent degradation benchmark
Wang *et al.* [34]	Operational constraints	CVaR	Energy + regulation	Risk-aware benchmark
Collath *et al.* [22], [23]	Calendar + cycle	Aging-aware scheduling	Intraday energy	Shows why aging representation can change dispatch
Aaltonen *et al.* [29]	Physics-based aging	Reinforcement learning	FCR	Connects bidding reward to aging assessment
Astero and Evens [30]	Battery operation	Control / technical requirements	FCR	Highlights operational requirements beyond capacity pricing
This work	**Occupancy + cycle**	**Mean–CVaR**	**DA + aFRR capacity**	**Controlled diagnosis of bid changes**

The fourth strand is risk-aware bidding. CVaR provides a tractable representation of lower-tail risk [31]–[33] and has been applied to VPP, PV-storage, and energy–frequency-regulation bidding [8], [34], [35]. Distributionally robust VPP models further extend uncertainty treatment [36].

The gap addressed here is narrower than any of these individual topics. The paper does not claim novelty in degradation, CVaR, stochastic bidding, or reserve participation separately. Rather, it isolates the incremental effect of *making SoC occupancy economically visible* while retaining cycle-related cost, upward reserve parking, and a common mean–CVaR market model. The empirical component is likewise framed as a diagnostic because reserve-capacity and imbalance series in the study package are proxies rather than day-specific German settlement records.

TABLE II
VPP, BATTERY, MARKET, AND RISK PARAMETERS USED IN THE NUMERICAL STUDY

Parameter	Value	Role
BESS power / energy	2.50 MW / 5.00 MWh	Study plant
PV / wind / load class	4 / 2 / 5 MW	Study plant
η_c / η_d	0.95 / 0.95	Efficiency
SoC range	10–90%	Operational constraint
DG capacity / cost	1.8 MW / 68 EUR/MWh	Dispatchable resource
DG ramp	1.2 MW/h	Operational constraint
Reserve cap	1.5 MW	Default sensitivity parameter
CVaR confidence α	0.95	Tail confidence
CVaR weight β	0.05	Risk weight
SoC bands N	8	Default ReLCCA resolution
Solver	HiGHS [37]	Open-source LP solver
Random seed	20260821	Reproducibility

Table I lists the closest studies. Most price either cycle depth or throughput. Only [28] combines calendar and cycle aging in a day-ahead energy–reserve model, and it uses deterministic inputs. None of the listed studies pairs SoC-dependent aging with a mean–CVaR bid, which is the gap this paper isolates.

III. SYSTEM, DATA, AND INFORMATION STRUCTURE

A. VPP Configuration

The VPP is modeled as a price-taking portfolio in the German DE-LU bidding area. It contains a 5 MW-class industrial load, 4 MW photovoltaic generation, 2 MW wind generation, a 2.5 MW/5 MWh stationary BESS, and a 1.8 MW dispatchable generator. The BESS has 0.95 charging and discharging efficiencies and operates within a 10–90% SoC range. A cyclic daily boundary condition is imposed. The generator has a marginal operating cost of 68 EUR/MWh and a 1.2 MW/h ramp limit. The default aggregate reserve cap is 1.5 MW. These values are fixed study parameters and are not fitted to the reported outcomes. Table II summarizes them.

B. Data and Scaling

The empirical panel covers 1 December 2025 00:00 through 21 August 2026 07:00 Berlin time and contains 6,319 hourly observations corresponding to 262 complete days. The panel has 263 calendar days. Of these, 262 are complete 24-hour days (6,288 h). The other 31 hours are 23 h on 29 March 2026 (daylight-saving change) and the first 8 h of 21 August 2026. All values are observed data downloaded on [date]; none are forecasts. Public

German market and physical series are obtained from Bundesnetzagentur SMARD [38] and the Fraunhofer ISE Energy-Charts interface [39]. The panel contains 403 negative-price hours. National physical series are mapped to the study portfolio with frozen scaling factors: the 95th-percentile load is mapped to 5 MW, while the study-window maxima of solar and wind are mapped to 4 MW and 2 MW, respectively. The factors are not re-estimated separately for individual days. Scenarios come from a daily block bootstrap. For an operating day D, we draw K vectors of forecast errors (load, PV, wind, price) from complete days before D, so the errors of one day move together. The panel starts on 1 December 2025, so autumn 2025 is not covered.

The information structure separates first-stage market commitments from scenario-dependent operating recourse. Day-ahead energy and reserve positions are common across scenarios, whereas charging, discharging, generator output, imbalance, and band-level operating variables respond to each scenario. Scenario probabilities are uniform within the daily scenario set.

C. Reserve and Imbalance Representation

The reserve-capacity prices are published German aFRR four-hour block means used in the study. Positive-direction means are 4.5, 12.0, 47.1, 18.4, 35.6, and 8.0 EUR/MW/h; negative-direction means are 6.0, 15.0, 36.9, 57.4, 20.0, and 2.4 EUR/MW/h [40], [41]. These values are held fixed across operating days. They should therefore be interpreted as a capacity-price proxy rather than as day-specific auction outcomes.

The imbalance component is represented by a documented two-price proxy based on the day-ahead price. For a short deviation, the settlement proxy is $\lambda^{\mathrm{DA}} + 18 + 0.12|\lambda^{\mathrm{DA}}|$; for a long deviation it is $\lambda^{\mathrm{DA}} - 10 - 0.08|\lambda^{\mathrm{DA}}|$, subject to minimum/maximum spreads of $+5$ and -3 EUR/MWh, respectively. A 150 EUR/MWh clipping case is evaluated only as a robustness test. Because directly mounted day-specific reBAP data were not available in the study package, the reported wholesale results are simulated profits under these settlement assumptions, not realized German settlement profits.

IV. Mathematical Formulation

A. Common Market Model

All three aging laws are embedded in the same daily optimization problem. Let t denote hour, ω scenario, and ℓ SoC band. The first-stage variables are the day-ahead energy position p_t^{DA} and reserve commitments $r_t^{\uparrow}$ and $r_t^{\downarrow}$. Recourse variables include battery charging $p_{t,\omega}^{\mathrm{ch}}$, discharging $p_{t,\omega}^{\mathrm{dis}}$, stored energy $E_{t,\omega}$, generator output $g_{t,\omega}$, net imbalance $z_{t,\omega}$, and ReLCCA band variables.

The physical balance for each hour and scenario is

$$p_t^{\mathrm{DA}} + z_{t,\omega} + p_{t,\omega}^{\mathrm{dis}} - p_{t,\omega}^{\mathrm{ch}} + g_{t,\omega} + W_{t,\omega} + PV_{t,\omega} = L_{t,\omega}. \tag{1}$$

Here p_t^{DA} is energy sold day-ahead (negative means bought) and z is energy bought in the imbalance market (negative means sold). At a negative price, a bought position earns money; this is what "long" means in the rest of the paper. The sign convention defines positive $z_{t,\omega}$ as a net deviation settled in the imbalance market. Renewable and generator limits are linear, and generator ramping is represented with linear intertemporal inequalities.

B. Battery Dynamics and Reserve Feasibility

The battery state evolves according to

$$E_{t,\omega} = E_{t-1,\omega} + \eta_c p_{t,\omega}^{\mathrm{ch}} \Delta t - \frac{p_{t,\omega}^{\mathrm{dis}}}{\eta_d} \Delta t, \tag{2}$$

$$E_{\min} \le E_{t,\omega} \le E_{\max}, \quad 0 \le p_{t,\omega}^{\mathrm{ch}}, p_{t,\omega}^{\mathrm{dis}} \le \overline{P}_B, \tag{3}$$

$$E_{t,\omega} = E_{\min} + \sum_{\ell} \Delta E_\ell u_{t,\omega,\ell}, \quad 0 \le u_{t,\omega,\ell} \le u_{t,\omega,\ell-1} \le 1, \tag{4}$$

$$E_{0,\omega} = E_{T,\omega}. \tag{5}$$

The cyclic condition is applied separately to each scenario. Upward reserve is constrained by both power capability and an energy-parking requirement. A generic form consistent with the implemented model is

$$r_t^{\uparrow} \le \overline{P}_B - p_{t,\omega}^{\mathrm{dis}}, \tag{6}$$

$$E_{t,\omega} \ge E_{\min} + \tau^{\uparrow} \frac{r_t^{\uparrow}}{\eta_d}, \tag{7}$$

$$r_t^{\uparrow} \le \overline{P}_B - p_{t,\omega}^{\mathrm{ch}}, \quad E_{t,\omega} + \eta_c \tau^{\uparrow} r_t^{\uparrow} \le E_{\max}, \tag{8}$$

where $\tau^{\uparrow}$ is the fixed reserve-sustainment parameter used by the study. The constraint prices the energy

that must remain available for upward deployment; it does not simulate the detailed activation trajectory or mileage of an aFRR signal.

No binary charging and discharging variable is introduced. The implemented parameterization and objective punish energy processing and do not reward simultaneous charging and discharging. This is an implementation assumption, not a general mathematical theorem. The LP claim reported here therefore applies only to the stated parameterization and not to arbitrary battery-market formulations.

C. Three Aging Representations

1) Throughput Model: The benchmark assigns a constant marginal aging cost to charge and discharge:

$$C^{\mathrm{THR}}_{t,\omega} = c_{\mathrm{thr}}\left(p^{\mathrm{ch}}_{t,\omega}+p^{\mathrm{dis}}_{t,\omega}\right)\Delta t, \quad c_{\mathrm{thr}} = 1.5 \text{ EUR/MWh}. \tag{9}$$

Idle stored energy is not directly priced by this term.

2) Daily DoD Model: The Xu benchmark represents daily depth by the normalized range of stored energy [16]. Let $E_{\max,d}$ and $E_{\min,d}$ be the maximum and minimum daily stored energy. The daily depth variable d_d satisfies

$$d_d \geq \frac{E_{\max,d} - E_{\min,d}}{E_B}, \tag{10}$$

and its cost is represented by a convex piecewise-linear epigraph with segment widths 0.20, 0.30, 0.30, and 0.20 and slopes 10, 40, 80, and 120 EUR per unit depth [16]:

$$d_d = \sum_{k=1}^{4} \delta_{d,k}, \quad C^{\mathrm{DoD}}_d = \sum_{k=1}^{4} m_k \delta_{d,k}, \tag{11}$$

$$\begin{aligned} 0 \leq \delta_{d,k} \leq w_k, \\ w = [0.2, 0.3, 0.3, 0.2], \\ m = [10, 40, 80, 120]. \end{aligned} \tag{12}$$

3) ReLCCA: The usable interval $[E_{\min}, E_{\max}]$ is divided into N equal-width SoC bands. Occupancy variables $u_{t,\omega,\ell}$ are represented cumulatively so that band filling is monotone. Band-level charge and discharge flows connect occupancy changes to battery movement. The ReLCCA aging cost is

$$\begin{aligned} C^{\mathrm{ReLCCA}}_{t,\omega} = &\sum_{\ell} \kappa_\ell u_{t,\omega,\ell} \Delta E_\ell \Delta t \\ &+ \sum_{\ell} \gamma^{\mathrm{ch}}_\ell p^{\mathrm{ch}}_{t,\omega,\ell} \Delta t \\ &+ \sum_{\ell} \gamma^{\mathrm{dis}}_\ell p^{\mathrm{dis}}_{t,\omega,\ell} \Delta t. \end{aligned} \tag{13}$$

The first term is the defining occupancy component: energy retained in a high-SoC band incurs a calendar-aging cost even when no energy is cycled during the hour. The second and third terms retain cycle-related costs. All coefficients are fixed parameters, so the formulation remains affine. Each κ_ℓ comes from

$$\kappa_\ell = \frac{\text{fade over ten months} \times 12/10 \times \text{replacement cost}}{8{,}760 \text{ h}}. \tag{14}$$

For the mid plateau this is $0.024 \times 1.2 \times 150{,}000/8{,}760 = 0.493$ EUR/MWh/h. The default eight-band κ map uses three plateaus derived from the Keil-based calibration [12]:

$$\kappa_\ell = \begin{cases} 0.247 \text{ EUR/MWh/h}, & \ell \in \mathcal{L}_{\mathrm{low}}, \\ 0.493 \text{ EUR/MWh/h}, & \ell \in \mathcal{L}_{\mathrm{mid}}, \\ 0.822 \text{ EUR/MWh/h}, & \ell \in \mathcal{L}_{\mathrm{high}}. \end{cases} \tag{15}$$

These values are an operational economic parameterization of published calendar-aging plateaus, not an independently fitted cell-aging model. The previously reported quadratic R^2 is not used and is intentionally omitted because the assigned plateaus are not independent observations.

D. Mean–CVaR Objective

For scenario ω, wholesale profit is represented as

$$\Pi_\omega = R_{\mathrm{DA},\omega} + R_{\mathrm{aFRR},\omega} - C_{\mathrm{imb},\omega} - C_{\mathrm{DG},\omega} - C_{\mathrm{age},\omega}. \tag{16}$$

A retail load-service overlay, when reported, is added after the wholesale optimization and therefore cannot change the market-decision ranking.

Let $L_\omega = -\Pi_\omega$ denote scenario loss. With confidence level $\alpha = 0.95$, the Rockafellar–Uryasev representation introduces a threshold ζ and nonnegative excess variables s_ω [32], [33]:

$$s_\omega \geq L_\omega - \zeta, \quad s_\omega \geq 0, \tag{17}$$

$$\mathrm{CVaR}_\alpha(L) = \zeta + \frac{1}{1-\alpha}\sum_{\omega} \pi_\omega s_\omega, \quad \alpha = 0.95. \tag{18}$$

The implemented objective maximizes expected profit while penalizing lower-tail loss:

$$\max \sum_{\omega} \pi_\omega \Pi_\omega - \beta\, \mathrm{CVaR}_\alpha(L), \quad \beta = 0.05. \tag{19}$$

All terms are affine under the stated assumptions. Hence the implemented problem is an LP. No

KKT reformulation, McCormick envelope, or Big-M construction is required. Piecewise linearization is used only for the daily DoD benchmark and the ReLCCA band representation; CVaR is linearized with auxiliary variables.

V. Experimental Design

A. Paired Identification Principle

Two bid types are compared throughout. B2 is the energy-only bid: the VPP sells day-ahead energy and no reserve. P1 is the joint bid: it sells day-ahead energy and aFRR reserve capacity. So P1$-$B2 is the extra money (or loss) that comes from adding reserve. The word "law" means the aging model used inside the optimization: throughput, Xu, or ReLCCA. The central experimental principle is paired comparison. For a given day, the plant configuration, market prices, scenario inputs, reserve schedule, risk parameters, solver, and numerical seed are held constant. Only the aging representation changes. This design minimizes confounding and permits the downstream market-position difference to be attributed to the aging formulation within the stated model. The first-stage bids (day-ahead energy and reserve) are fixed from the scenario solution. They are then settled on the realized day; the second-stage variables are re-solved with the same aging law while the bids stay fixed. All profits in Tables III–VI come from this settlement.

B. Isolation Days

Four isolation dates are used to represent ordinary and stressed conditions: 20 January, 5 April, 1 May, and 19 August. The default scenario count is $K = 30$. These dates are diagnostic examples rather than a representative random sample.

C. Counterfactual Negative-Price Intervention

To test whether negative-price depth alone is sufficient to reproduce the observed reversal, the 19 August residual portfolio is frozen while the selected midday price interval is replaced by scaled copies of the 1 May trough. The injected minimum is varied over -50, -80, -200, -300, -414, and -499 EUR/MWh. The experiment therefore changes price severity while preserving the underlying residual structure; it is a counterfactual intervention, not a market forecast.

D. Historical Negative-Price Scan

The complete panel is scanned for days with minimum day-ahead price below -20 EUR/MWh. The scan contains 20 qualifying days; Table V reports the nine days below -50 EUR/MWh. The analysis is used to assess the frequency of ranking reversal within the available study window, not to estimate a population-level probability.

E. Profit Decomposition and Robustness

Whenever aging formulations produce materially different market positions, wholesale profit is decomposed into day-ahead energy, reserve, imbalance, generator, and aging components. Robustness checks vary the number of SoC bands, calendar coefficient mapping, calendar/cycle components, scenario count, calendar annualization, imbalance clipping, reserve cap, and battery energy capacity. These tests evaluate sign stability within the model and do not constitute electrochemical validation.

VI. Results

A. Isolation-Day Comparison

Table III compares the three aging models on four days. Each row is a paired run, so only the aging model changes. In the last column, a positive number means that adding reserve (P1) earned more than the energy-only bid (B2). On 20 January, 5 April, and 19 August the sign is positive for all three models. On 1 May it turns negative for throughput and Xu but stays positive for ReLCCA.

The isolation results do not show a systematic dominance of ReLCCA. On 20 January, 5 April, and 19 August, reserve participation improves the reported simulated wholesale outcome under all three aging laws. The main distinction occurs on 1 May, where the throughput and Xu energy-only cases move into a different day-ahead position while ReLCCA does not.

B. Controlled Price-Trough Experiment

Fig. 1 pushes the midday price lower and lower on the 19 August portfolio. Both curves fall slowly but stay well above zero, even at -499 EUR/MWh (throughput from $+1{,}145$ to $+790$ EUR, ReLCCA from $+1{,}353$ to $+1{,}056$ EUR). So a deeper trough alone does not recreate the 1 May reversal.

TABLE III
PAIRED ISOLATION-DAY RESULTS UNDER THE THREE BATTERY-AGING REPRESENTATIONS ($K = 30$, 180 EUR/MWH LOAD-SERVICE OVERLAY INCLUDED; DIFFERENCES ARE COMPUTED FROM UNROUNDED VALUES)

Day	Law	B2 (EUR)	P1 (EUR)	P1−B2 (EUR)
20 Jan	Throughput	10,330	11,260	+930
	Xu	10,332	11,258	+926
	ReLCCA	9,886	11,100	+1,214
5 Apr	Throughput	15,475	16,676	+1,201
	Xu	15,475	16,676	+1,201
	ReLCCA	15,215	16,603	+1,389
1 May	Throughput	14,492	14,246	−246
	Xu	14,492	14,225	−266
	ReLCCA	13,180	13,696	+515
19 Aug	Throughput	10,961	12,268	+1,307
	Xu	10,961	12,268	+1,306
	ReLCCA	10,766	12,245	+1,480

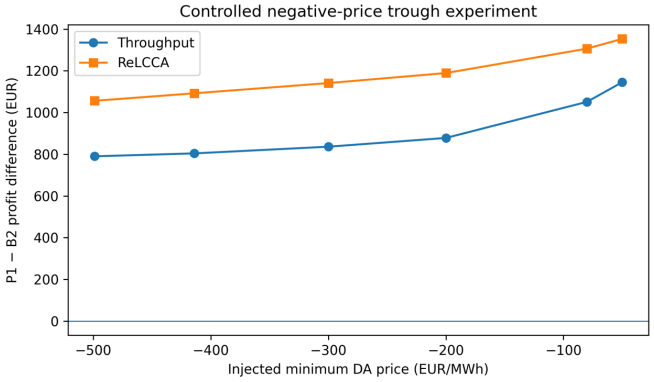


Fig. 1. Controlled negative-price trough experiment using the 19 August residual structure. The intervention varies only the injected midday price minimum; it is not a forecast.

The throughput and ReLCCA differentials remain positive throughout the intervention range. Thus, making the trough progressively more negative does not, by itself, reproduce the 1 May reversal. This result supports a conditional interpretation: price severity interacts with the residual portfolio state and the feasible value of stored energy.

C. 1 May Profit Decomposition

Table IV splits each result into its parts. The key comparison is B2-Throughput against B2-ReLCCA: the day-ahead energy line differs by 1,246 EUR, while the aging line differs by only 4 EUR. In other words, the aging model changed the bid, not the aging bill. In both P1 rows, aFRR capacity pays between 1.3 and 1.5 kEUR. Fig. 2 shows the same numbers as bars; the tall bar for B2-Throughput (+978 EUR) has no counterpart in B2-ReLCCA (−268 EUR).

TABLE IV
1 MAY 2026 WHOLESALE PROFIT DECOMPOSITION FOR THE ENERGY-ONLY BOOKS ($K = 30$, NO OVERLAY)

Model	DA energy	aFRR	Imbalance	DG	Aging	Wholesale
B2–Throughput	+978	0	+600	−1,714	−58	−194
P1–Throughput	−541	+1,333	+436	−1,635	−32	−440
B2–ReLCCA	−268	0	+462	−1,647	−54	−1,506
P1–ReLCCA	−1,238	+1,457	+474	−1,635	−49	−991

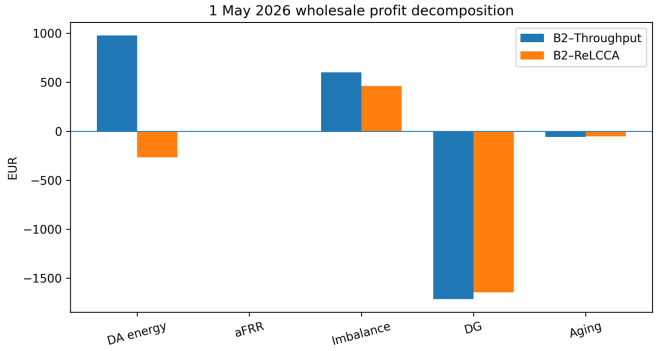


Fig. 2. 1 May 2026 wholesale profit decomposition for the two energy-only books.

The central comparison is between B2-Throughput and B2-ReLCCA. Their day-ahead energy components differ by 1,246 EUR (+978 versus −268 EUR), whereas their direct aging costs differ by only 4 EUR (−58 versus −54 EUR). The resulting wholesale-profit difference is 1,312 EUR (−194 versus −1,506 EUR). The result therefore cannot be explained by a large direct degradation bill. Rather, the aging representation changes the optimization trade-off and the resulting day-ahead position. This statement is a consequence of the decomposition; it does not by itself establish a causal shadow-price estimate.

D. Historical Negative-Price Scan

Table V checks whether 1 May is special. Among the nine deepest days, throughput is negative only on 1 May (−246 EUR). 26 April is the closest call: the price fell to −414 EUR/MWh, yet throughput still gains +139 EUR. ReLCCA is positive on all nine days. The other 11 days with a minimum between −20 and −50 EUR/MWh are positive for both models. Each row is a different day with a

TABLE V
HISTORICAL NEGATIVE-PRICE DAYS BELOW −50 EUR/MWH AND CORRESPONDING MODEL DIFFERENTIALS ($K = 20$)

Day	Minimum DA price	Throughput P1−B2	Xu P1−B2	ReLCCA P1−B2
1 May	−499	−246	−251	+531
26 Apr	−414	+139	+137	+869
25 Apr	−191	+812	+850	+1,114
2 May	−130	+887	+887	+1,188
6 Apr	−126	+1,043	+1,043	+1,382
5 Apr	−105	+1,182	+1,182	+1,381
11 Apr	−80	+1,100	+1,100	+1,246
24 May	−76	+1,115	+1,115	+1,316
29 Apr	−60	+1,074	+1,060	+1,273

TABLE VI
ROBUSTNESS AND SENSITIVITY CHECKS FOR THE RELCCA P1−B2 DIFFERENTIAL

Check	ReLCCA P1−B2 (EUR)
$N = 4/6/8/12$, $K = 20$	+565 / +533 / +576 / +561
κ step / all-mid / all-high, $K = 30$	+515 / +514 / +512
Calendar only / cycle only / both, $K = 30$	+587 / +512 / +515
$K = 20$ / $K = 30$	+576 / +515
$\sqrt{t}$ annualization ($\kappa \times 0.913$)	+516
Imbalance $\lvert\lambda_{\mathrm{DA}}\rvert$ clipped at 150	+541 (throughput −232)
$r^{\max} = 0.5$ / 2.5 MW, $K = 20$	+234 / +800 (throughput −91 / −262)
$E_b = 2$ MWh, $K = 20$	+409 (throughput +19)

different portfolio state, so this is a count in our data, not a probability.

The full scan contains 20 days below −20 EUR/MWh; the table above shows the nine days below −50 EUR/MWh. Throughput is negative only on 1 May within the full qualifying set, while ReLCCA remains positive on all 20 qualifying days. Because the sample contains only 20 qualifying days and the period is incomplete, this should be interpreted as an in-sample diagnostic frequency rather than a statistical estimate of the probability of reversal.

E. Numerical and Parameter Robustness

Table VI reports the robustness checks. Fig. 3 shows how the ReLCCA differential changes with the number of SoC bands N; the value remains stable and positive. Both the calendar-only and the cycle-only variants keep P1−B2 positive (+587 and +512 EUR). So it is the dependence of the cost on the SoC level, not the calendar term alone, that changes the bid. Models that are blind to the SoC level (throughput and daily depth) do not.

The ReLCCA differential retains a positive sign across the tested band resolutions and alternative parameterizations. The checks therefore indicate numerical stability within the implemented model,

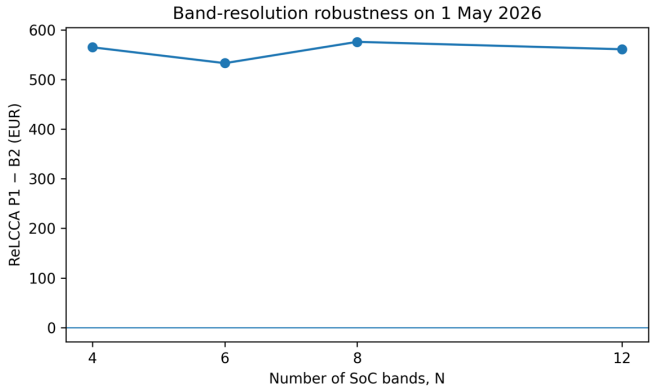


Fig. 3. Band-resolution robustness on 1 May 2026. The ReLCCA differential remains positive and stable across different numbers of SoC bands.

while making no claim of electrochemical validation.

VII. DISCUSSION

A. Interpretation of the Market-Position Mechanism

Throughput and daily-depth formulations assign aging cost primarily to processed energy or to the magnitude of the SoC excursion. ReLCCA introduces a further dimension: the location of stored energy within the SoC range itself carries an aging cost. Once upward reserve is committed, a portion of that stored energy must remain available for deployment. Under stressed price conditions, the optimizer consequently weighs the value of changing the energy position against both occupancy exposure and reserve feasibility.

The 1 May decomposition makes the distinction particularly clear. A direct aging-cost difference of only 4 EUR coincides with a 1,246 EUR difference in the day-ahead energy component. The correct interpretation is therefore that the aging representation modifies objective coefficients sufficiently to change the optimal active set and, with it, the market position. It does not mean that the aging bill itself generates a 1,246 EUR saving.

B. Post-Optimization CMRI Diagnostic

The Capacity Margin and Risk Index (CMRI) is retained solely as an ex-post descriptor of portfolio state. It is neither a decision variable nor a mecha-

nism for changing constraints or switching between optimization formulations:

$$\mathrm{CMRI}_t = \frac{\mathrm{SoC}_t - \mathrm{SoC}_{\min}}{\mathrm{SoC}_{\max} - \mathrm{SoC}_{\min}} - \gamma \frac{R_t^{\uparrow} + R_t^{\downarrow}}{P^{\mathrm{rated}}} - \delta\, \psi(\pi_t^{\mathrm{DA}}), \tag{20}$$

$$\psi(\pi_t^{\mathrm{DA}}) = \max\left(0, -\frac{\pi_t^{\mathrm{DA}}}{\pi_{\mathrm{base}}}\right) \exp(\beta_p \mathrm{SoC}_t). \tag{21}$$

The first term measures normalized energy headroom, the second the reserve-obligation burden, and the third negative-price stress. The index is descriptive and may be used to characterize the portfolio state associated with bid differences. It is deliberately excluded from the optimization to preserve the LP formulation and avoid an unsupported threshold-switching mechanism.

C. Scope of the Empirical Evidence

The evidence does not establish universal superiority of ReLCCA, a universal negative-price threshold, a population-level probability of reversal, or a validated cell-level degradation law. Because reserve and imbalance inputs are proxy representations, the monetary outcomes are simulated wholesale profits under the stated settlement assumptions rather than realized German market settlements.

D. Practical Implications

For a BESS owner who bids energy and reserve jointly, the results carry a narrow but actionable message. When midday prices fall sharply negative, a throughput-based model may value stored energy mainly through the cost of cycling it. If the same battery must also hold energy for upward reserve, a throughput model can therefore underestimate the economic cost of parking that energy at a high SoC. Under the right residual portfolio state, this omission is large enough to flip the preferred day-ahead position. The practical recommendation is not to abandon throughput models, which remain simple and robust on ordinary days, but to treat SoC-occupancy pricing as a stress-case override. Operators who observe both deep negative prices and binding reserve commitments should re-run the bid with an occupancy-aware term before submitting the day-ahead schedule.

VIII. Limitations and Reproducibility

First, the diagnosed condition occurs once within the 262 complete days, and September–November are absent from the study panel. The observed frequency cannot therefore be generalized to a full-year population. Second, ReLCCA is an optimization-oriented economic encoding of published aging relationships rather than a cell-level electrochemical model; its κ coefficients require independent calibration for the relevant chemistry, temperature, and operating history.

Third, the baseline calendar annualization is linear in time and does not reproduce the exact square-root dependence used in some aging formulations. The $\sqrt{t}$ experiment is a robustness check rather than an electrochemical replacement. Fourth, the 150 EUR/kWh replacement value is a cell-replacement economic assumption rather than installed EPC cost. Fifth, aFRR prices are repeated four-hour block means rather than day-specific cleared auction records, and activation-energy trajectories are not simulated.

Sixth, imbalance prices are generated by a documented two-price proxy rather than directly observed reBAP settlements. Seventh, the Xu benchmark is a daily-depth piecewise-linear representation and is not equivalent to rainflow cycle counting. Eighth, the VPP is a price taker and does not model network constraints, strategic price making, or market power. Finally, the event scan is based on an incomplete study period and should be reproduced from the preserved configuration rather than manually transcribed.

The implementation uses Python/Pyomo with HiGHS [37] and the stated random seed. The preserved project package contains the model source, data, statistics, and reproducibility notes. Future work should replace market proxies with day-specific settlement records, extend the data window to a complete calendar year or longer, estimate uncertainty across a larger event sample, and calibrate occupancy-aging coefficients against measured cell or field-BESS data.

IX. Conclusion

This paper developed a controlled diagnostic for determining when explicit SoC-occupancy aging can change a risk-aware BESS energy–reserve decision. Three aging representations were embedded in

a common linear mean–CVaR market model, allowing the aging representation to be changed without changing the surrounding market architecture. On ordinary isolation days, the formulations generally preserve the same qualitative value of reserve participation. On 1 May 2026, however, the throughput energy-only formulation becomes long during a -499 EUR/MWh midday trough, whereas ReLCCA remains short. The corresponding 1,312 EUR difference in wholesale profit is dominated by a 1,246 EUR change in the day-ahead energy component, while the direct aging-cost difference is only 4 EUR.

The controlled trough intervention and historical negative-price scan indicate that the reversal is not explained by negative-price depth alone. The residual portfolio state also matters. The principal methodological contribution is a transparent separation between direct aging expenditure and the downstream market-position response induced by the aging representation. The study should not be interpreted as validation of a universal degradation law or as a reconstruction of realized German aFRR settlement. Its practical implication is narrower: when a BESS is simultaneously exposed to energy-price stress and reserve-related SoC requirements, explicitly pricing SoC occupancy can materially change the selected market position relative to formulations based only on throughput or daily depth.

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